

South Africa liquid fuels market story

Inland supply, corridor capture and tank
turnover

Durban-Lesedi corridor into Gauteng

7 October 2026

Executive summary: baseline, changes and investment outlook;
calibration gaps explicit

Photo source: Royal Vopak | Terminal Durban | vopak.com



Slow demand growth and greater import reliance set the baseline; power, refineries and routes could change it

Baseline / reference	What the evidence establishes	What changes the outlook
 Fuel demand and power	2013–2023 CAGR: petrol –2.08%; diesel +0.82%. Eskom OCGT fuel use fell 40% in FY2025.	Growth, rail, EVs and efficiency alter transport demand; fleet recovery and new generation alter diesel dispatch.
 Domestic supply and imports	2020–2025 imports rose by 5.43bn litres of diesel and 2.73bn of petrol. The sourcing mix has shifted.	SAPREF and PetroSA restart paths, Secunda gas/MRG and Natref output can change domestic supply.
 Corridor position and tank turnover	Durban and Lesedi provide coastal and inland assets serving Gauteng. Working capacity and turns are open.	Route costs, access and customer capture determine flow; achievable turnover determines handling headroom.
 Medium / Medium reference world	M/M anchors the nine-world matrix. Historical evidence is assembled; numerical calibration remains open.	Vary demand and supply around M/M, then test capture, turnover and returns to identify investment triggers.

Executive summary — baseline and changes. Comparable historical periods differ: Department sales 2013–2023; SARS imports 2020–2025; Eskom FY2025 vs FY2024 (own diesel + kerosene). M/M is a reference framework, not a calibrated forecast.

Vopak can capture inland growth through the Durban–Lesedi corridor, with expansion conditional on turnover and returns

Investment outlook	Why it matters	What determines the choice
 Capture inland supply	Imported and domestic fuel can serve inland demand. Terminal access and customer capture determine volumes.	Size the accessible market by origin, product and destination, then estimate Vopak's share of the resulting flows.
 Use the corridor position	Durban and Lesedi connect coastal handling with inland distribution into the Gauteng catchment.	Compare delivered costs, access and reliability with competing gateways and domestic sources.
 Increase asset turnover	Growth can increase throughput through existing tanks. Receipt and dispatch constrain achievable turns.	Use existing assets, debottleneck or add storage according to working capacity, turnover and returns.
 Test nine market worlds	Demand and domestic supply each have L/M/H paths. Medium/Medium is the reference to calibrate.	Track captured flow, required turns and capacity headroom in every world; phase investment against clear triggers.

Working thesis for the Durban–Lesedi corridor into Gauteng. Domestic and imported origins require verified access. No measured Vopak share, calibrated nine-world outputs or approved expansion returns are asserted.

Capturable volumes, route competitiveness and tank turnover determine the investment required

Argument	Situation	Complication	Resolution / decision
Transport demand	Subdued aggregate growth masks different product trends. Road freight supports diesel use.	Sector growth, rail recovery, EVs and efficiency change fuel demand.	D1–D4 define activity, rail, EV and efficiency paths. Translate them into capturable product demand.
Power demand	Power shortages supported diesel use; Eskom consumption has recently fallen.	Recovery may persist or reverse through outages and replacement delays.	D5 adds fleet recovery, replacement timing and diesel dispatch to the same demand paths.
Domestic supply	Reduced local production has increased reliance on imports.	SAPREF, Natref and Secunda can change domestic fuel output materially.	S1–S4 define plant output and sourcing mix. Imports are one consequence of the domestic balance.
Corridor competition	Durban connects imported fuel to inland customers.	Alternative gateways, delivered costs, reliability and access alter flows.	T1–T2 compare delivered economics and capture for domestic and imported fuel serving common destinations.
Vopak investment	Durban and Lesedi provide connected assets whose throughput can rise through higher turnover.	Working tankage, receipt/dispatch, peaks and stock requirements constrain achievable turnover.	T3–T5 compare working capacity, required/achievable turns and dispatch limits. F1 tests incremental returns.

Lever IDs are mapped in the appendix. All nine market worlds use explicit routing, capture and operational assumptions. Tank occupancy, turnover and throughput utilisation are separate measures.

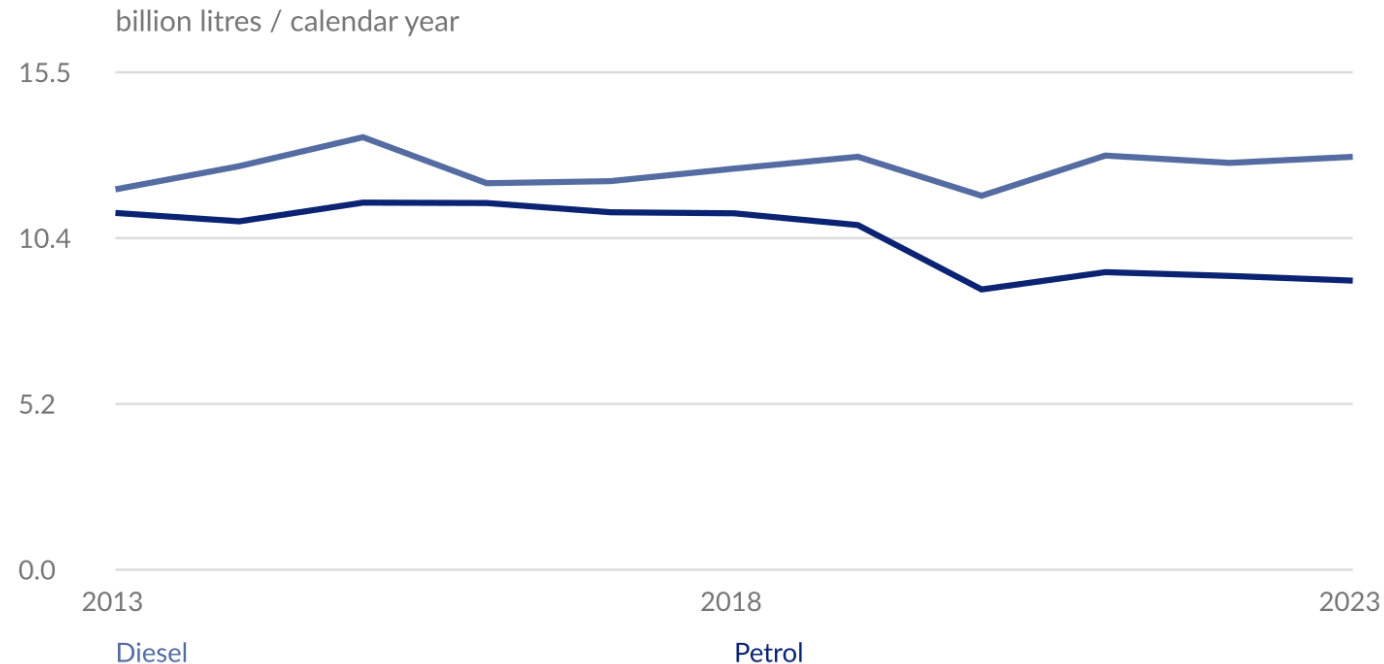
Nine demand–supply worlds test the investment choice against the Medium / Medium baseline

Fuel demand / domestic supply	HIGH supply (S1–S4)	MEDIUM supply (S1–S4)	LOW supply (S1–S4)
LOW demand D1–D5	L / H LOWEST IMPORT PRESSURE Domestic routing and customer capture still determine terminal turns.	L / M Lower imports than baseline. Test utilisation and customer capture.	L / L Demand and supply both fall. Net import effect depends on their relative change.
MEDIUM demand D1–D5	M / H Lower imports than baseline. Supply recovery reduces import needs.	M / M BASELINE Reference sourcing, captured throughput, required turns and headroom.	M / L Higher imports than baseline. Supply losses increase import needs.
HIGH demand D1–D5	H / H Demand and supply both rise. Net import effect depends on their relative change.	H / M Higher imports than baseline. Demand growth increases import needs.	H / L HIGHEST IMPORT PRESSURE Test the captured flow against working capacity and achievable turns.

Every cell: sourcing mix → captured Durban / Lesedi flow → required vs achievable turns → headroom → return.

Qualitative matrix, not numerical forecasts. T1–T5 and F1 translate each world into investment outcomes. Imports are an intermediate balance; domestic supply can also generate terminal flows where access is available.

Petrol demand has declined while diesel has grown slowly, limiting the case for demand-led expansion



Petrol 2013–2023

11.15 to 9.04 bn litres

-2.12 bn L / -19.0%

CAGR -2.1% per year

Diesel 2013–2023

11.89 to 12.91 bn litres

+1.02 bn L / +8.6%

CAGR +0.8% per year

FIASA 2024 observations are excluded from this comparison: source definitions are not reconciled.

The demand history supports a selective investment case: growth must translate into captured corridor throughput and sufficient tank turns.

Department sales 2013–2023. CAGR from comparable endpoints: petrol -2.08%, diesel +0.82%. FIASA 2024 is excluded from the trend calculation; its observations remain in the evidence data.

Power shortages and road freight have supported diesel consumption, but that support can change

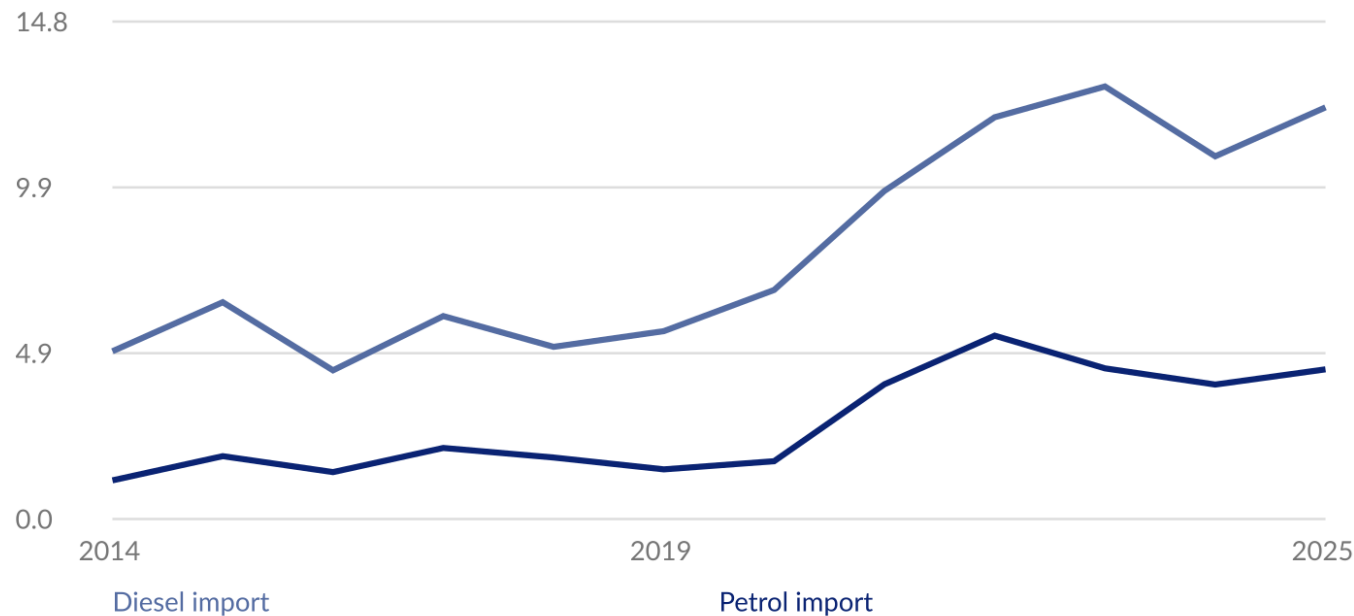
Demand support	Historical evidence	What this establishes
Power generation	Eskom's own OCGT fuel use rose to 1,129.5m litres in FY2024, then fell to 679.1m in FY2025 as EAF improved.	D5: availability, retirement and new generation determine dispatch. See the quantitative power exhibit and calibration requirements.
Road freight	The national payload series shows road carrying substantially more tonnage than rail.	D2: transferable tonne-kilometres, rail reliability and fuel intensity determine the net diesel effect.
Demand interpretation	Mining and agriculture add sector-specific diesel exposure; aviation has a separate jet-fuel market.	D1: sector activity drives end-use demand. Histories precede explicit L/M/H assumptions; missing data remain identified.

Eskom 2025 integrated report (own-fleet diesel + kerosene); Stats SA freight payload. No causal diesel attribution from tonnes alone. Details: appendix pp19, 22–23.

Lower domestic production has increased imports and shifted how inland demand is supplied

IMPORT TREND

billion litres / calendar year



CHANGE SINCE 2020

Diesel import

6.81 to 12.25 bn litres

Increase: 5.43 bn litres

CAGR: 12.4% (2020–2025)

Petrol import

1.72 to 4.45 bn litres

Increase: 2.73 bn litres

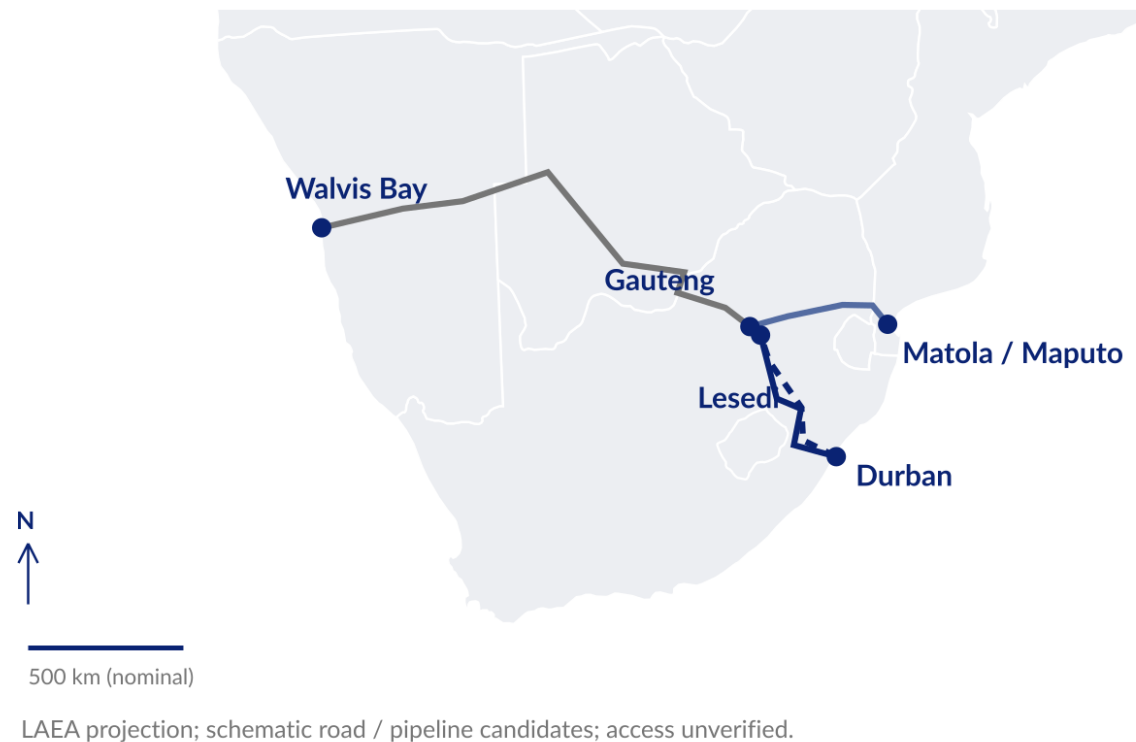
CAGR: 21.0% (2020–2025)

More inland handling opportunity;
capture and turnover determine benefit.

Additional imported litres create handling demand. Vopak benefits where it captures inland flows and can increase turnover or profitably remove a constraint.

SARS litre-reported petrol/diesel codes, complete calendar years. 2020–2025 changes are historical comparisons, not plant-level causal attribution. Export histories retained in the source report; unit/blend exclusions remain.

Domestic sources and competing gateways contest the same inland customers



SAME PRODUCT, SAME DESTINATION

Delivered cost = product + handling +
transport + border + losses + finance.

Bearable logistics charge = destination
price less product, other costs and margin.

Headroom = bearable less actual charge.
Capacity and reliability constrain capture.

Domestic origins include Secunda / Natref;
routes, access and tariffs need validation.

Compare the same product at the same Gauteng destination. Price headroom only becomes capturable throughput where capacity, rights and reliable access exist.

Natural Earth country geometry; shared LAEA projection with uniform scale. GeoJSON source catalogue: output/delivered/geospatial/2026_10_07. Corridor lines and places remain authored approximations; operating routes, facility positions and access require validation.

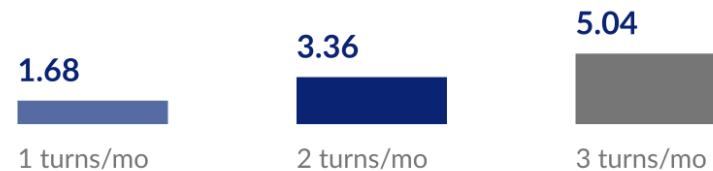
Higher turnover could let Durban and Lesedi serve more demand through existing tanks

GROSS HANDLING SENSITIVITY

Durban



Lesedi



INVESTMENT TEST

Required turns = captured annual throughput
÷ eligible working capacity ÷ 12.

Compare with achievable receipt / dispatch,
pipeline access, peaks and stock policy.

Existing assets first; debottleneck or add
tanks where a profitable constraint remains.

bn litres/year; 1 / 2 / 3 monthly turns are an existing authored sensitivity, not measured site limits.

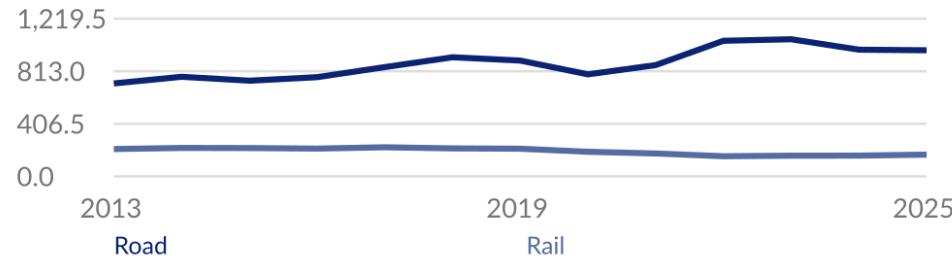
Required working-capacity turns must be compared with achievable turns at each terminal. Gross handling equivalents do not establish spare capacity or actual market share.

Registered REG-TERMINAL-HANDLING-TURNS: 1/2/3 turns per month, authored operational sensitivity, not the demand-supply worlds or measured limits. Gross Durban mixed-product tankage; no assumed eligible fraction. Do not sum inter-terminal transfers as unique demand.

Rail recovery, EV adoption and efficiency could reduce transport fuel demand

Rail recovery

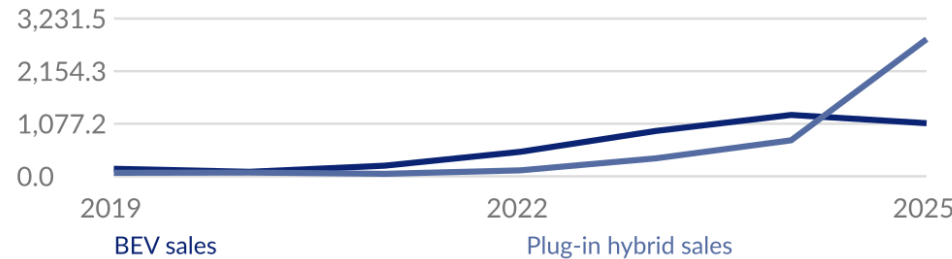
million tonnes



D2: capacity, reliability, transferable freight
Tonnes require haul distance and traction
before conversion to net diesel savings.

EV uptake

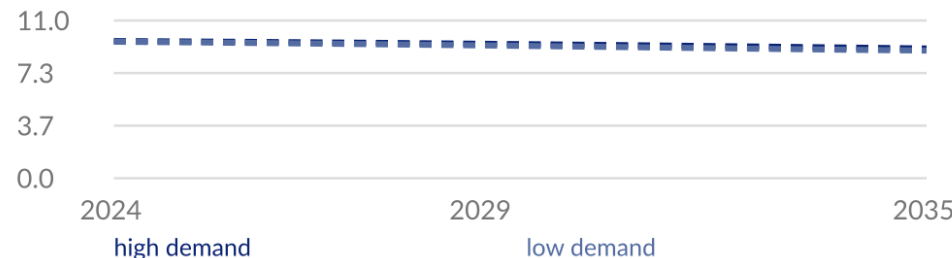
vehicles sold / year



D3: affordability, charging, fleet replacement
Sales history is not fleet penetration.
L/M/H adoption calibration remains open.

Fleet efficiency

L/100 km; draft engine

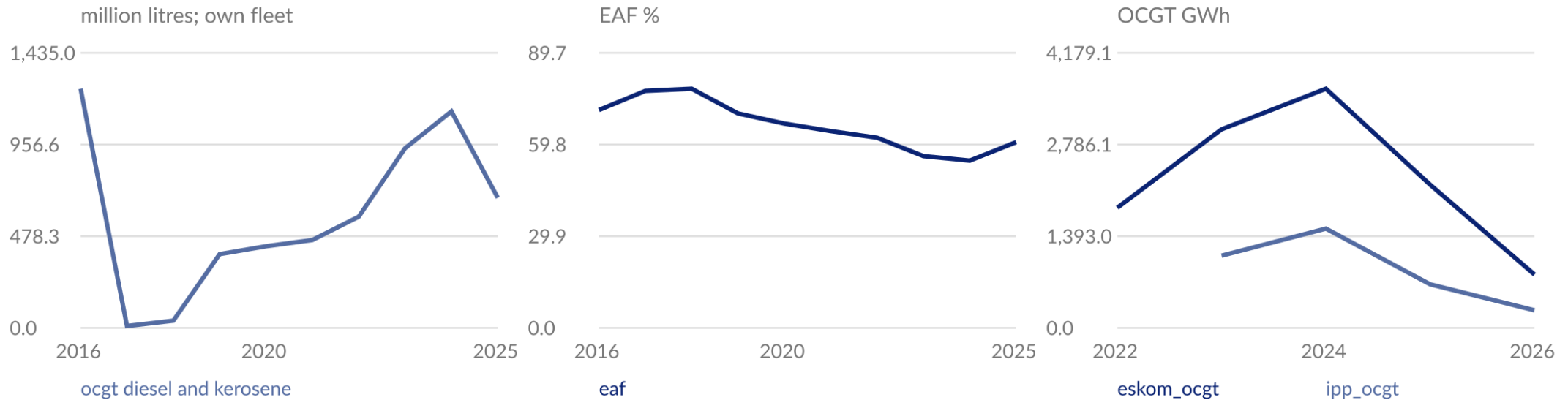


D4: new-cohort efficiency and mileage
Dashed paths are existing model diagnostics.
OEM history and Medium calibration are open.

The time series establish the starting points. Rail, adoption and efficiency levers must be calibrated into the same three demand paths before the matrix is quantified.

Stats SA freight payload; staged naamsa BEV/PHEV annual sales 2019–2025; existing engine petrol-ICE efficiency diagnostics. Dashed efficiency paths are assumptions, not historical OEM evidence. Medium and causal fuel impacts remain uncalibrated.

Power recovery could lower diesel use further, while outages and replacement delays could reverse the gains



D5 levers: fleet availability, retirement dates, wind / solar / gas commissioning and diesel dispatch.

Next quantitative layer: demand and generation by technology (GWh), then residual diesel litres.

Load-shedding / private-backup history and calibrated replacement-generation paths remain open.

Eskom recovery has lowered diesel dispatch. Wind, solar and gas timing must be tested against demand, fleet retirements and dependable output before forecasting residual diesel.

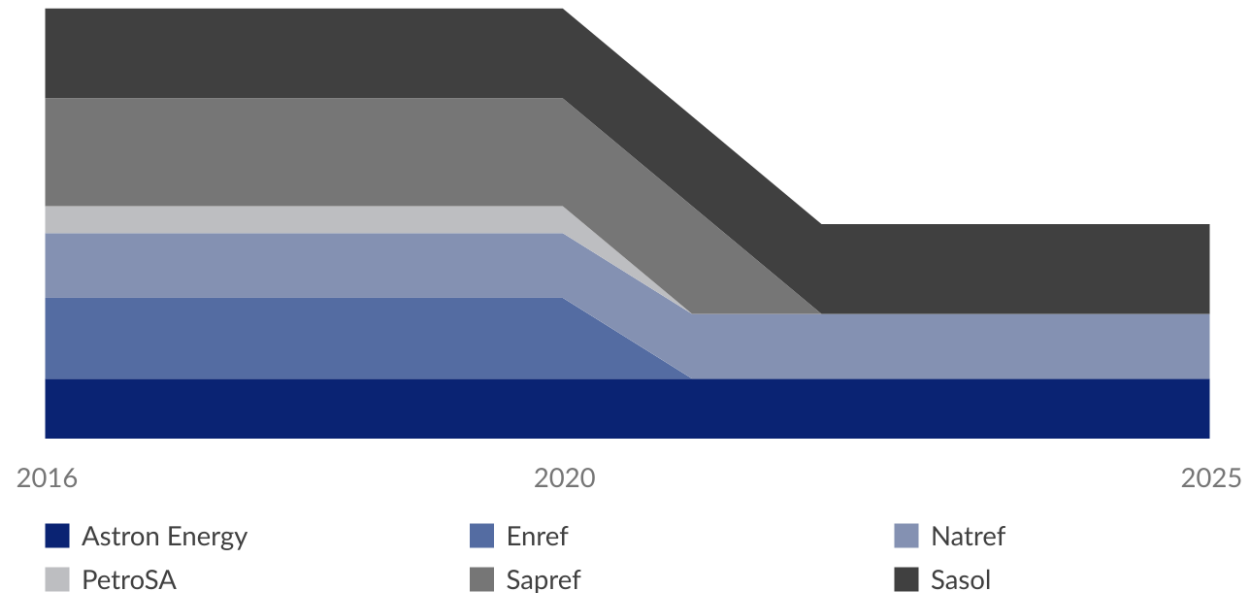
Eskom FY ending March. Fuel/EAF: 2025 report; GWh: staged 2024–2026 reports. Own Eskom and IPP generation separated. Private backup and load-shedding histories are missing; no assumed annual generation gap is assigned entirely to diesel.

Refinery recovery could reduce import needs, while further production losses could increase them

REPORTED CAPACITY BY PLANT

thousand bbl/day; all products

718 in 2016 → 358 in 2025



Historical capacity is not actual production. L/M/H output paths are not yet calibrated.

SUPPLY-WORLD LEVERS

LOW: slower restarts / output losses

MEDIUM: reference plant paths

HIGH: recovery / additional output

S1 SAPREF: restart, ramp and yield

S2 Secunda: gas availability / MRG

S3 Natref: availability and utilisation

S4 PetroSA: restart and liquid yields

The existing stacked chart explains the capacity footprint. Plant availability, utilisation and product yields are still required to turn L/M/H supply assumptions into fuel-output paths.

FIASA 2025 p49, staged_refinery_capacity_reported.csv. Reported nameplate capacity, not actual production; source dashes excluded from totals. Historical stack reused; no invented Medium or L/H production forecasts. SAPREF/Natref/Secunda/PetroSA remain explicit levers.

Changes in delivered cost and reliability could shift inland flows between competing corridors

Evidence	Interpretation	Implication for Vopak
Durban–Lesedi	Compare road, rail and pipeline to the same receiving point and product specification.	T1–T2: compare access, cost and reliability to the same destination; quantify captured throughput and the resulting tank turns.
Matola / Maputo	Include inland haul and border costs for each reachable destination.	A port-level cost comparison cannot establish Gauteng competitiveness.
Walvis Bay	Test distance, transit time and corridor capacity into each target market.	Do not assume that every inland customer is economically addressable.
Price bearability	Bearable logistics charge = destination selling price less origin product cost, other costs and required margin.	Headroom = bearable charge less actual logistics cost. Capacity and service constrain the share captured through Durban–Lesedi.

Submodel specification adopted from the annotated review. No route winner is asserted without matched costs, access and capacity.

Expansion requires capturable throughput beyond achievable capacity at an adequate return

Investment test	Required measure	Decision consequence
Captured throughput	Domestic + imported unique deliveries by product, customer and terminal.	Establish accessible flows and avoid double-counting Durban–Lesedi transfers.
Required turnover	Annual captured m ³ / eligible working m ³ / 12 = required monthly turns.	Compare with achievable turns, seasonality, stock policy and receipt/dispatch limits.
Capacity headroom	Achievable annual handling less required throughput; tank occupancy assessed separately.	Use existing capacity where available, then test targeted debottlenecking or additional storage.
Financial return	Storage and handling revenue, operating cost, capex and working capital.	Test incremental cash flow, break-even utilisation, NPV and IRR before committing to expansion.

Public capacity ratios and Gauteng market size are screening evidence only. Working capacity, realised dispatch, routing, tariffs, contracts and capex remain open. No investment return or throughput market share is fabricated.

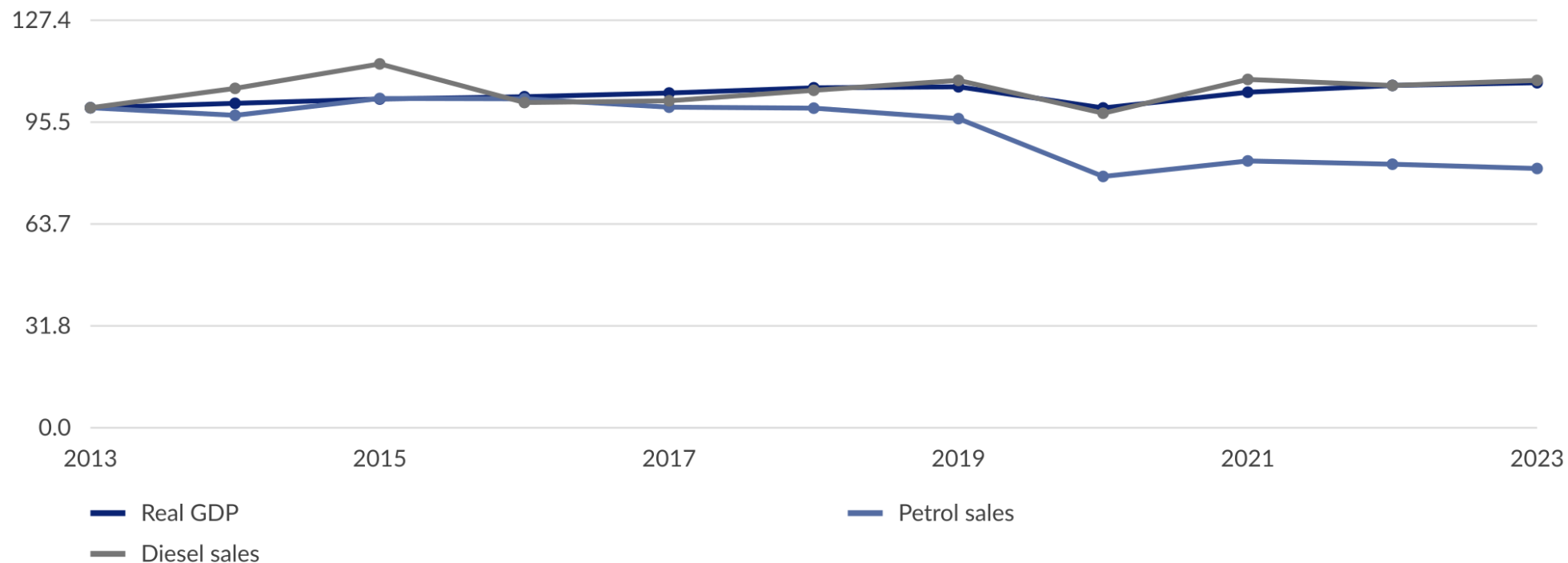
The supporting evidence calibrates demand, supply and the terminal investment tests

Evidence group	Detail retained	Connection to the story
Demand	Activity, freight, EVs, efficiency and power	D1–D5 create the three demand paths.
Supply	Plant output, gas availability and restart timing	S1–S4 create the three domestic-supply paths.
Capture and operations	Routes, customer share, working tankage and turnover	T1–T5 translate each world into terminal flows and constraints.
Returns	Rates, costs, capex and funding	F1 translates each investment option into cash flow and returns.

The main story runs from the opening thesis to the investment tests. The added lever register and power/supply calibration pages record what remains to be populated.

Fuel demand has diverged from GDP, so sector activity and fuel intensity need separate treatment

index: 2013 = 100; matched 2013-2023 window

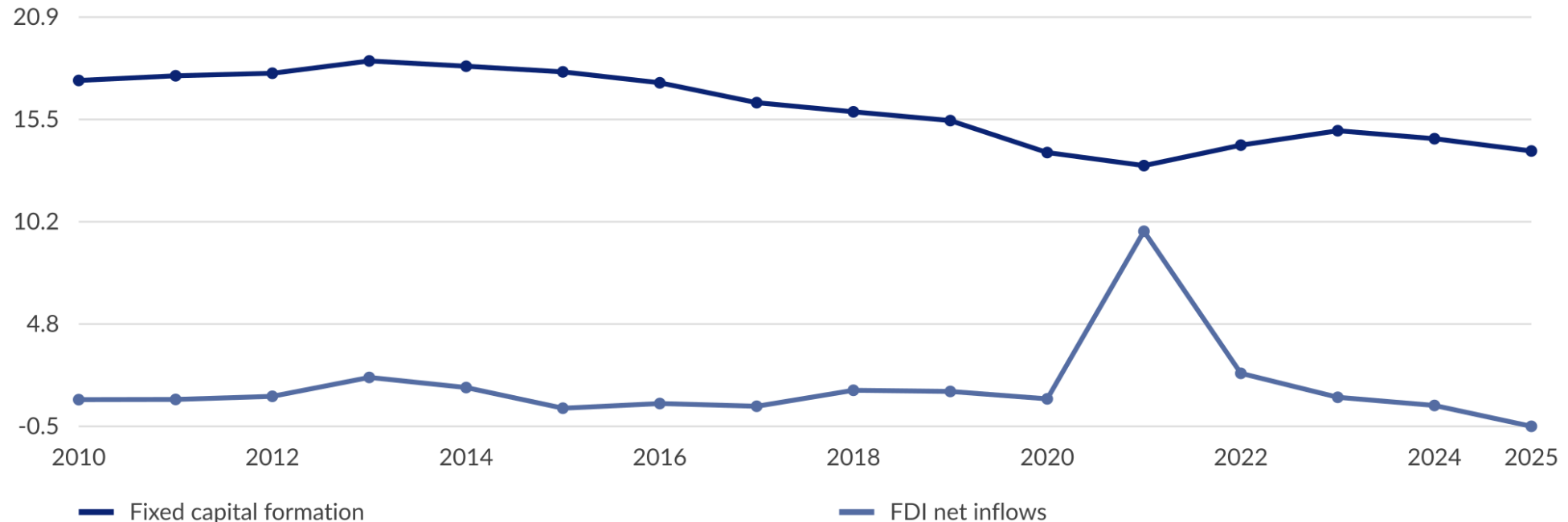


Economic weakness means limited growth in real activity and investment. Structural change means a different mix of output, travel and fuel use; this chart alone cannot attribute causality.

Staged GDP and department fuel sales, indices 2013=100. Sector attribution remains open; COVID distorts short-period comparisons.

Fixed investment has weakened while FDI fluctuates, limiting a simple growth-led fuel argument

percent of GDP / calendar year



Fixed investment helps frame productive capacity. FDI can include large transactions; neither series is a direct investor-confidence measure or an additional fuel-demand multiplier.

World Bank WDI, FDI net inflows and gross fixed capital formation as % of GDP, 2010–2025. Confidence series remains an evidence gap.

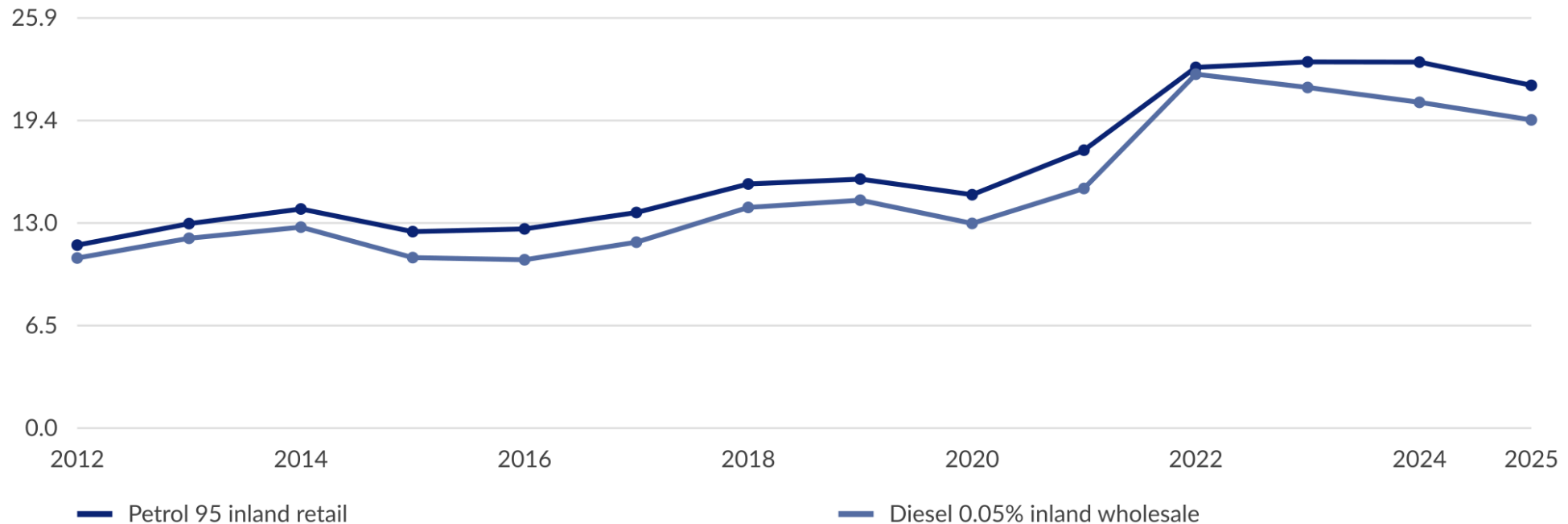
Mining, agriculture and aviation expose fuel demand to different activity drivers

Evidence	Interpretation	Implication for Vopak
Mining	Output, commodity mix and haul distance influence diesel use and freight.	Separate mine activity from the fuel impact of changing freight mode.
Agriculture	Crop cycles, weather, irrigation and logistics can shift seasonal diesel use.	Establish sector activity and fuel exposure before assigning a headwind.
Aviation	Traffic, distance, load factors and aircraft efficiency drive jet-fuel demand.	Keep jet fuel separate from petrol/diesel shares and verify terminal eligibility.
Manufacturing	Real output and industrial composition help explain the economic backdrop.	Test changes using sector data; a national fuel chart does not prove deindustrialisation.

Analytical framework following user comments. Sector-specific exhibits and causal attribution remain to be completed.

Higher fuel prices strengthen EV running-cost incentives, but ownership economics determine uptake

nominal ZAR/litre; full-year monthly averages



Shipping disruption, crude prices and FX can move fuel costs temporarily. EV adoption depends on sustained total ownership cost, purchase affordability, charging access and fleet replacement.

Department fuel-price history, nominal R/litre. No quantified geopolitical premium or calibrated EV adoption response is asserted.

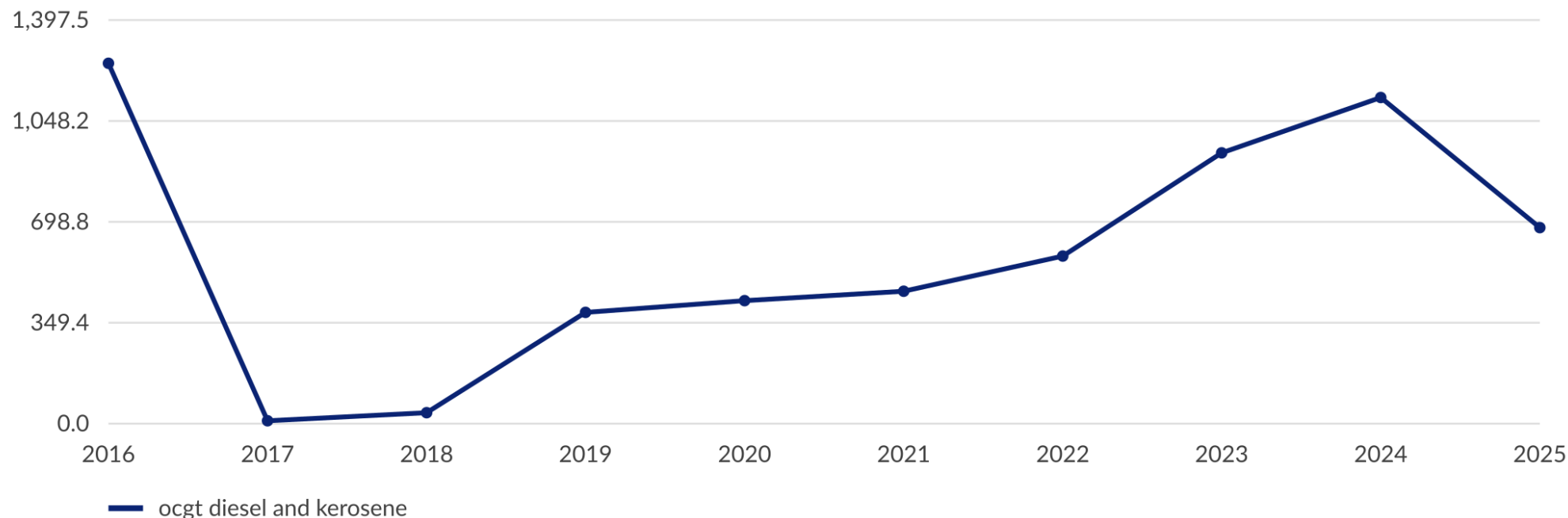
New-vehicle efficiency gains affect fuel demand gradually as older vehicles leave the fleet

Evidence	Interpretation	Implication for Vopak
Fleet growth	More vehicles or distance travelled can raise demand.	Measure activity before attributing lower fuel volumes to efficiency.
Efficiency	New-cohort consumption gains enter the fleet over successive replacement cycles.	Obtain OEM and real-world evidence by segment; do not apply new-vehicle gains to every vehicle.
EV substitution	EVs remove some combustion-engine vehicle activity from the fuel calculation.	Count each vehicle transition once, then calculate efficiency on the remaining combustion fleet.

Mechanism supported by existing cohort architecture. OEM historical and future efficiency evidence is still missing; no new forecast shown.

Eskom's own OCGT fuel use fell 40% in FY2025 as availability improved

million litres / FY ending 31 March

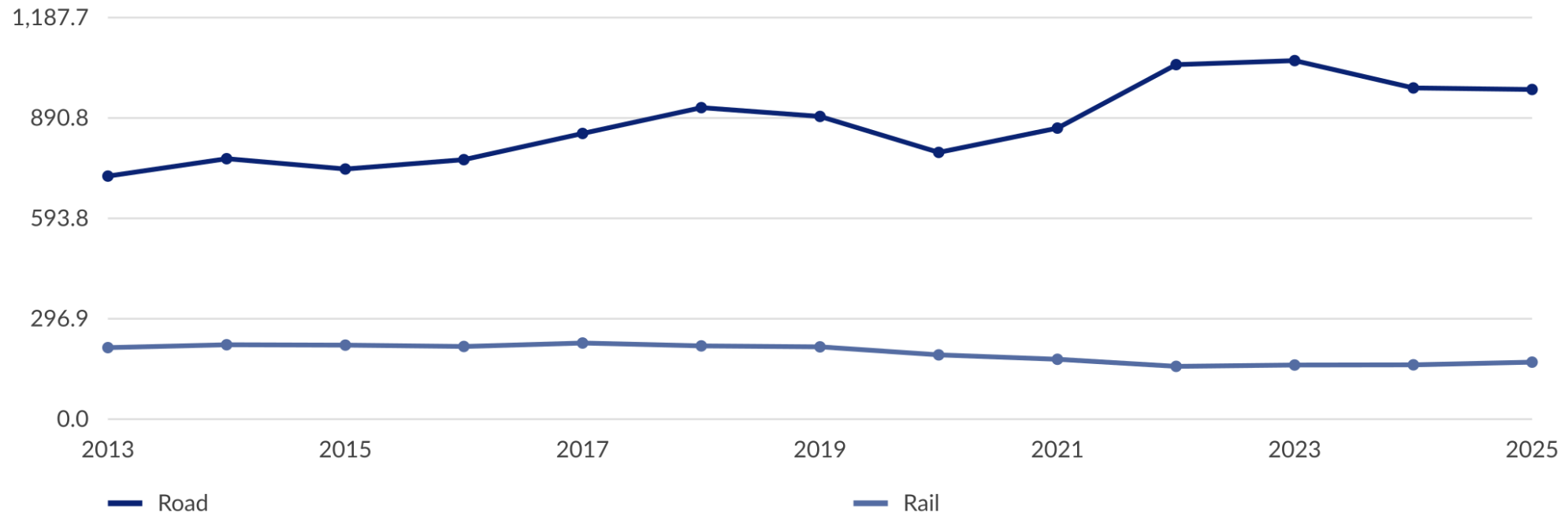


Reported fuel fell from 1,129.5 to 679.1 million litres as EAF improved from 54.56% to 60.60%. This removes part of the recent diesel-demand support.

Eskom integrated report 2025: FY March, own-fleet diesel + kerosene. EAF association is not a dispatch model; private backup use excluded.

Road carries most reported freight tonnage, making transferable haul distance a key diesel lever

million tonnes / complete calendar year



Payload establishes the historical direction. Diesel displacement requires corridor tonne-kilometres, road fuel intensity, rail traction and last-mile activity.

Stats SA monthly freight payload, complete calendar years. National tonnes are not corridor tonne-kilometres or transferable freight.

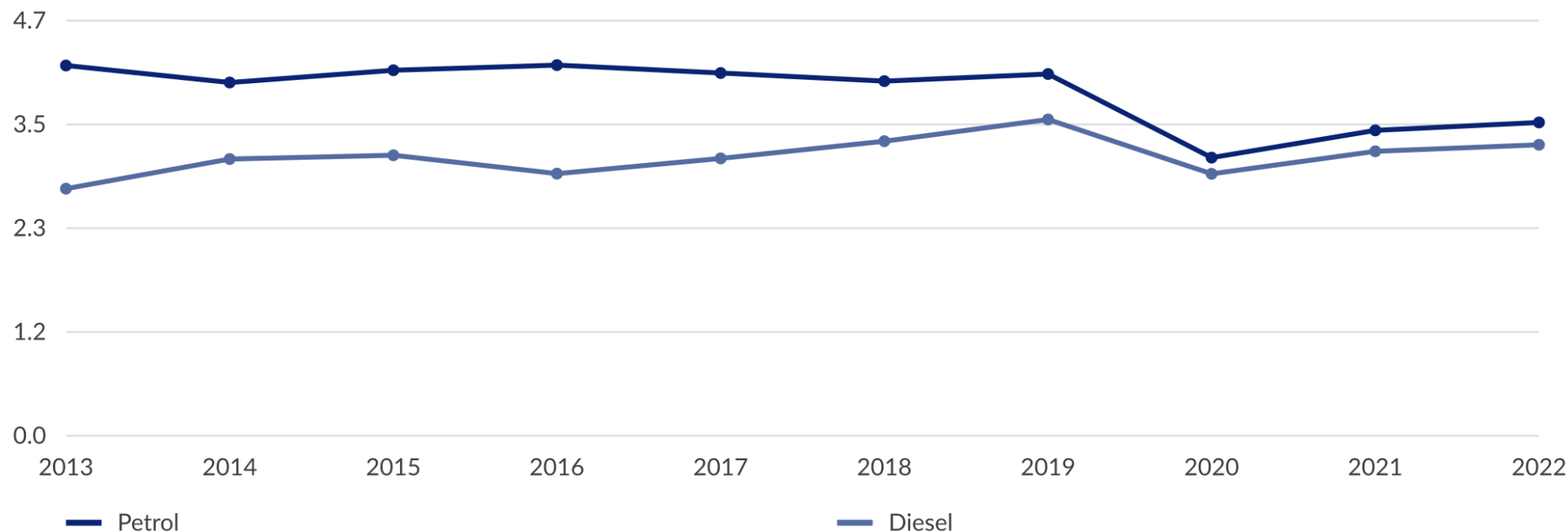
Rail funding and reliable service determine how much road freight can return to rail

Evidence	Interpretation	Implication for Vopak
Network ambition	The supplied brief reports a 250 Mt target dependent on a funding plan.	Do not treat every additional rail tonne as road displacement. Commodity, corridor and origin matter.
Funding constraint	Brief: R46.34bn approved over five years, including R1.71bn expansion.	Maintenance and sustaining spend dominate. A published plan is insufficient evidence of delivered capacity.
Container corridor	Brief: R2.11bn in 2026/27–2028/29; private funding expected thereafter.	Prioritise Durban–Gauteng road/rail competition and reconcile the corridor to Network Statement v4.
Two fuel effects	Rail can displace truck diesel while improved infrastructure also changes fuel-distribution economics.	Keep freight-demand effects separate from the route delivering fuel to Gauteng.

User-supplied AIA Transnet Funding Brief, Sep 2026, pp1–2; cites FY2026 AFS and Network Statement v4 Annexure 17. Primary figures not independently reconciled.

Gauteng demand establishes market scale, while Lesedi capture and turnover determine throughput

billion litres / complete calendar year



2022 petrol + diesel sales total 6.81bn litres. Lesedi's 140,000m³ equals 2.06% of that annual market per effective gross-capacity turn. Actual turns and Gauteng allocation remain unknown.

Department provincial sales, latest complete staged year 2022; Vopak Lesedi current capacity. Cross-vintage scale test only, not current observed market share.

Every market world passes through explicit demand, supply, capture and operating levers

Lever IDs	Inputs to calibrate	Output / remaining evidence
D1–D5 demand	D1 sector growth; D2 rail; D3 EV adoption; D4 efficiency; D5 power.	Historical volumes and time series feed L/M/H demand. OEM, corridor tonne-km and backup evidence remain open.
S1–S4 supply	S1 SAPREF; S2 Secunda gas/MRG; S3 Natref; S4 PetroSA.	Dates, utilisation, availability and yields produce L/M/H domestic output. Medium and plant marginal effects require calibration.
T1–T2 capture	T1 delivered route economics/access; T2 customer/product/destination capture.	Allocate domestic and imported fuel to unique terminal deliveries. Tariffs, rights and commercial allocation remain open.
T3–T5 operations	T3 eligible working tankage; T4 turnover; T5 receipt/dispatch, peaks and stock policy.	Compare required with achievable turns, occupancy and headroom. Existing 1/2/3-turn sensitivity is illustrative only.
F1 finance	Storage/handling terms, costs, capex, tax, working capital and hurdle rate.	Translate constrained throughput into cash flow, NPV, IRR and investment triggers. Commercial inputs remain open.

Medium is the reference market world; operational sensitivities are separate. Owner: Nigel / analyst and commercial teams. Resolve each missing input before numeric adoption or investment reliance.

Generation timing must close the power balance before residual diesel can be forecast

Time series	Assumption / lever	Evidence status and next use
Recovery history	EAF, unserved energy / load-shedding hours, OCGT GWh and litres.	EAF, Eskom fuel and Eskom/IPP GWh staged. Comparable load-shedding and private backup series still needed.
Generation supply by year	Existing fleet and retirement profile; wind, solar and gas commissioning; availability and utilisation.	Obtain dated Eskom / approved planning schedules. Report annual GWh alongside dependable capacity and dispatch constraints.
Residual dispatch	Electricity demand, hourly/seasonal shape, storage, imports, reserve needs and fuel constraints.	An annual GWh shortfall alone cannot determine diesel use. Calibrate feasible peaking dispatch, then apply consumption rates.
L/M/H demand contribution	Sustained recovery, reference and setback assumptions across the same planning horizon.	Output generation and diesel time series with solid actuals / dashed assumptions. No uncalibrated future values are drawn.

Requested generation-balance exhibit remains quantitatively incomplete. Owner: Nigel / power analyst; trigger: before D5 adoption. Do not convert installed renewable MW directly into dependable diesel displacement.

Plant-level output and yields must turn the capacity stack into three supply paths

Plant lever	Required time-series inputs	Scenario treatment
S1 SAPREF	Restart/FID dates, ramp-up, product yields and operating availability.	Distinguish import operations, announced refinery ambition and commissioned output.
S2 Secunda	Actual liquid output, gas decline, MRG allocation and marginal liquids yield.	Quantify output lost or retained; gas diversion is not an end-user demand reduction.
S3 Natref	Plant output and ownership basis, outages, utilisation and product yields.	Reconcile reported share with whole-site output before aggregation.
S4 PetroSA	Restart scope, timing, feedstock, utilisation and liquid yields.	Separate stopped GTL history from a funded restart assumption.

Three aligned L/M/H production panels require these inputs. Existing stacked history is nameplate capacity. Owner: Nigel / supply analyst; resolve before adopting nine-world domestic supply outputs.

Public capacity ratios establish scale but cannot substitute for realised terminal throughput

Catchment	Initial inference	What the result can support
Durban	45.1% of the combined published gross capacity of Vopak and Bidvest in Durban.	A two-operator footprint proxy with mixed products and an incomplete market denominator. It does not establish Durban throughput share.
Gauteng / Lesedi	2022 petrol + diesel sales: 6.81bn litres. Lesedi's gross capacity equals 2.06% of that market per effective annual turn.	A cross-vintage scale test. Current sales, eligible working capacity, actual turns and Gauteng allocation are needed for a throughput estimate.
Investment implication	Convert eligible tankage and realised turns into unique customer deliveries, then compare with existing capacity.	A large market can coexist with spare terminal capacity. Market scale alone does not establish a need for additional tanks.

Operator capacities: Vopak Durban 360,246m³, Bidvest Durban 439,306m³, Lesedi 140,000m³. Department Gauteng sales 2022; appendix p25. Transfers between terminals must not be double-counted.